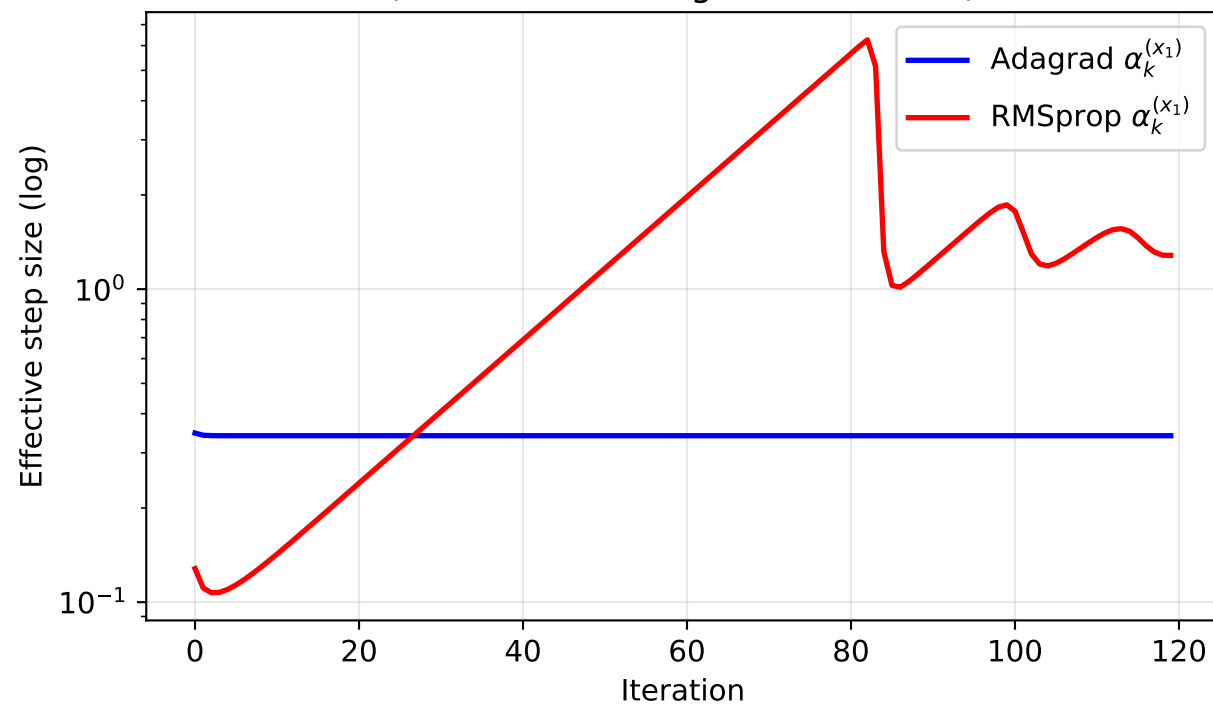
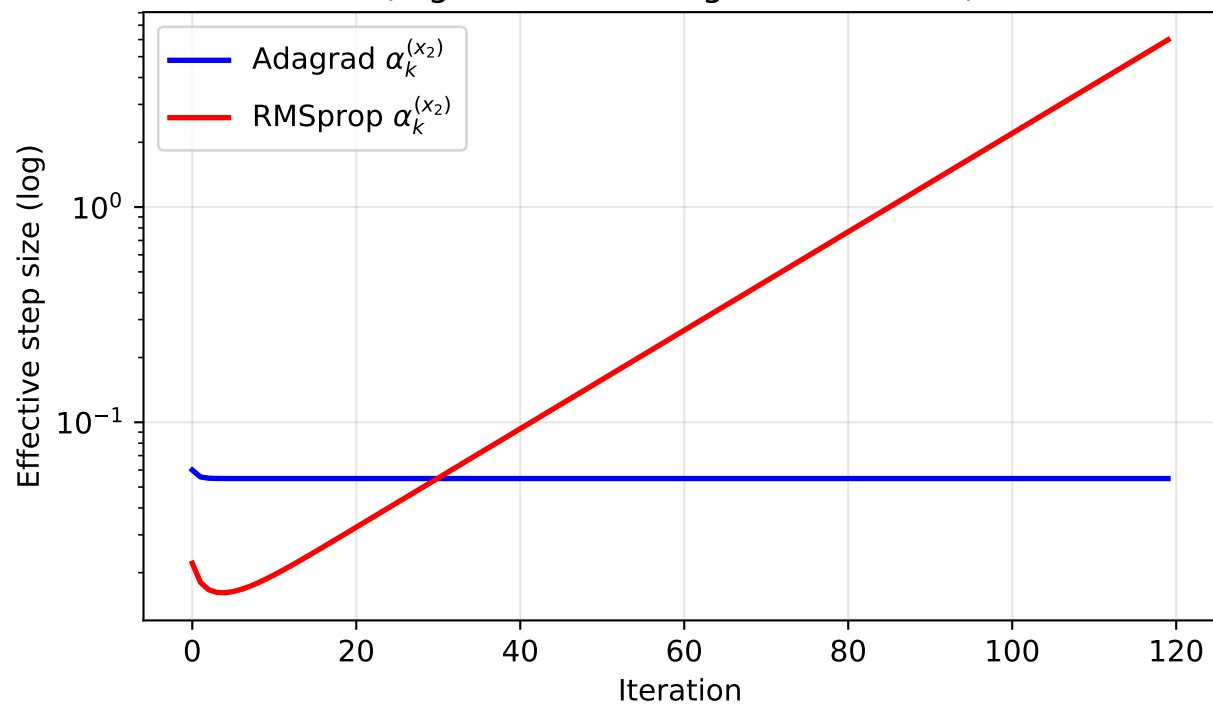


Q1: Adagrad vs RMSprop — Coordinate-wise Effective Step Size Analysis

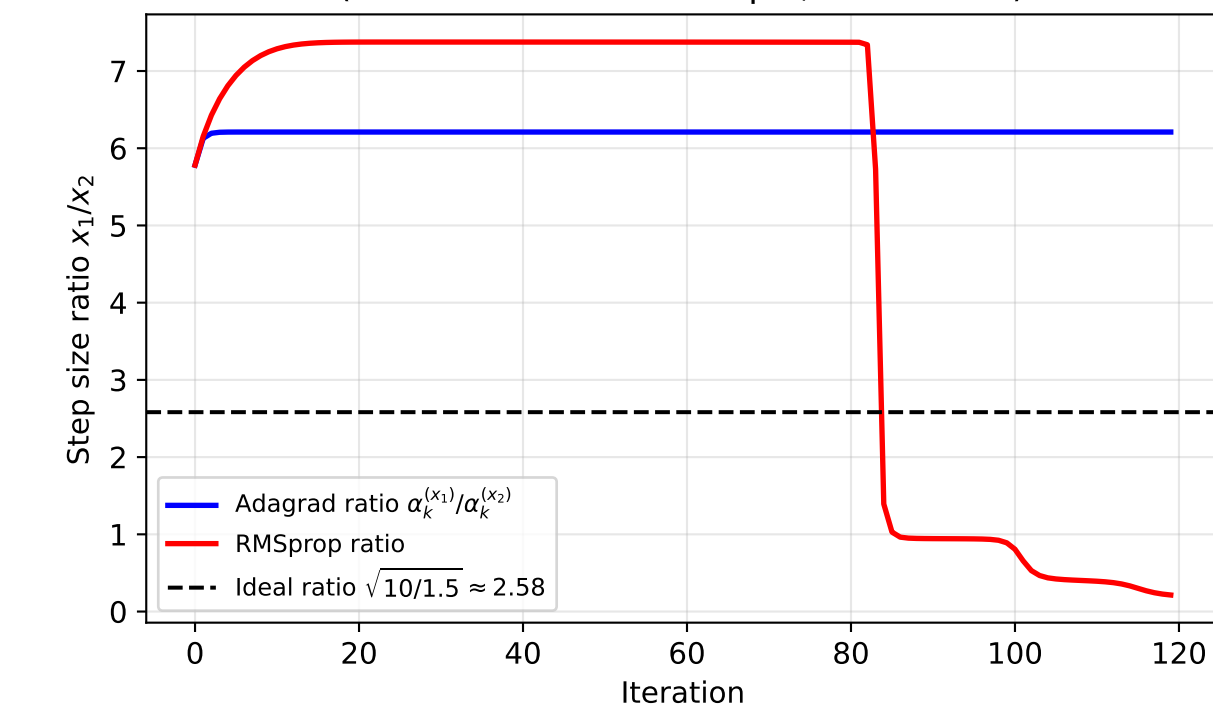
Effective Step Size: x_1 direction
(lower Hessian eigenvalue ≈ 1.5)



Effective Step Size: x_2 direction
(higher Hessian eigenvalue = 10)



Adaptive Preconditioning: Step Ratio
(ratio = 1 means isotropic, ideal = $\sqrt{\kappa}$)



Convergence on Benchmark B
(Adagrad reaches $f^* = 0.7244$ monotonically)

